From Detection to Decision: A Video-Level, Risk-Aware Pothole Prioritization Framework Using YOLOv8-N

ABSTRACT

Road infrastructure monitoring is a major challenge, especially in developing regions where high traffic, weather conditions, and limited maintenance accelerate road deterioration. Potholes are a common and dangerous form of damage, leading to vehicle damage, congestion, and accidents. Traditional inspection methods based on manual surveys or sensors are costly, time-consuming, and difficult to scale.

This paper proposes a video-based intelligent road condition assessment framework using YOLOv8-Nano for automated pothole detection without specialized hardware or IoT sensors. The system processes road videos by extracting representative frames, detecting potholes, and analysing their severity, degradation, and risk using visual features and morphological growth modelling. A temporal aggregation strategy converts frame-level detections into a video-level maintenance priority score, enabling practical road-level decision-making. Experiments on the RDD2022 dataset demonstrate reliable detection accuracy and explainable prioritization, making the framework scalable and suitable for intelligent transportation and infrastructure management systems.

KEYWORDS

Pothole Detection, Road Condition Monitoring, Video Analytics, Deep Learning, YOLOv8-Nano, Intelligent Transportation Systems.

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1.INTRODUCTION

Efficient road condition monitoring plays a vital role in ensuring traffic safety, reducing vehicle operating costs, and supporting sustainable transportation infrastructure. With rapid urbanization and increased vehicle ownership, road networks are subjected to heavy traffic loads and environmental stress, leading to frequent surface defects such as potholes. These defects not only degrade ride comfort but also significantly increase the risk of accidents and infrastructure maintenance costs.

Traditional road inspection techniques, including manual surveys and complaint-based reporting systems, are labour-intensive, subjective, and incapable of providing timely and consistent assessments at scale. In recent years, computer vision and deep learning techniques have emerged as promising solutions for automated pothole detection. Convolutional Neural Networks (CNNs) and object detection models such as YOLO and Faster R-CNN have demonstrated strong performance in identifying potholes from images and videos.

However, most existing studies focus on frame-level detection accuracy and do not address the broader challenge of road-level maintenance prioritization. Moreover, sensor-based approaches using accelerometers, GPS, or LiDAR increase system cost and deployment complexity, limiting scalability. To address these limitations, this work proposes a video-based pothole prioritization system that transforms raw road videos into actionable maintenance decisions using lightweight deep learning models and temporal risk aggregation.

2.PROBLEM STATEMENT

Road maintenance agencies are required to continuously monitor road conditions and prioritize repair activities to ensure safety and efficient transportation. However, existing road inspection practices largely depend on manual surveys or complaint-based reporting systems, which are time-consuming, subjective, and incapable of providing consistent assessments at scale. Although recent deep learning–based approaches have demonstrated promising results in automated pothole detection using images and videos, most existing systems operate at the frame-level or image-level, producing isolated detection outputs that do not translate into road-level maintenance priorities.

Furthermore, many pothole detection frameworks rely on sensor-based or hardware-dependent solutions, such as accelerometers, LiDAR, or embedded edge

devices, which increase deployment cost and restrict scalability across large road networks. Existing vision-based methods primarily focus on detection accuracy while neglecting pothole severity estimation, degradation prediction, and temporal aggregation, all of which are critical for realistic maintenance planning. As a result, road authorities are left with raw detection outputs that lack contextual risk interpretation and actionable insights.

In addition, most current systems function as black-box models, offering limited explainability and no integrated decision-support mechanisms to assist engineers or policymakers. The absence of a unified, software-only framework that can analyse road inspection videos, assess pothole severity and future degradation, and convert frame-level detections into video-level priority scores represents a significant research and practical gap. Addressing this gap is essential for developing scalable, cost-effective, and intelligent road condition assessment systems suitable for real-world deployment.

3.LITERATURE REVIEW

3.1 Traditional Approaches for Pothole Detection

Early research on pothole detection relied heavily on manual inspection and rule-based image processing techniques such as edge detection, thresholding, and contour analysis [15]. While these methods offered low computational complexity, they were highly sensitive to lighting conditions, shadows, and road texture variations.

Sensor-based approaches using accelerometers, gyroscopes, vibration sensors, and LiDAR systems mounted on vehicles were also explored to detect pothole-induced disturbances [4], [17]. Although these systems achieved reasonable accuracy, they required specialized hardware, frequent calibration, and increased deployment cost, making them unsuitable for large-scale road monitoring [11].

3.2 Deep Learning–Based Image-Level Pothole Detection

With the advancement of convolutional neural networks (CNNs), pothole detection shifted toward image-based deep learning models. CNN architectures enabled automatic feature extraction from road images, significantly improving detection robustness under varying environmental conditions [13]. Public datasets such as the Road Damage Dataset (RDD) facilitated standardized evaluation of these models [10].

Maeda et al. demonstrated that CNN-based models could reliably detect potholes and cracks using RDD, achieving higher accuracy than handcrafted features [10].

However, these models treated each frame independently and did not provide information about pothole severity, degradation, or maintenance urgency.

3.3 YOLO-Based Real-Time Pothole Detection Models

To meet real-time requirements, researchers adopted YOLO-based single-stage object detection frameworks. Several studies reported high detection accuracy and real-time performance using YOLO variants trained on RDD and custom datasets [2], [7], [8].

YOLOv8 demonstrated improved efficiency and accuracy due to architectural enhancements; however, most YOLO-based works focused only on bounding box detection without incorporating severity estimation or temporal aggregation [9], [12]. Additionally, many implementations relied on embedded or edge hardware, increasing system complexity [7].

3.4 Severity Estimation and Degradation Analysis

Recent studies emphasized pothole severity estimation using geometric features such as area, texture roughness, and depth approximation [9], [14]. However, depth estimation typically required stereo cameras or LiDAR, limiting scalability.

Degradation modelling, which predicts pothole progression over time, remains largely underexplored. Existing approaches often rely on historical maintenance records or sensor data, making them unsuitable for vision-only systems [18]. Consequently, degradation-aware prioritization remains a major research gap.

3.5 Video-Based and Temporal Analysis in Road Assessment

Although road inspections are commonly performed using videos, most studies process videos as independent frames, ignoring temporal consistency [8]. Limited research has explored temporal aggregation, and even fewer studies translate temporal information into actionable maintenance priorities [12]. This disconnects limits real-world applicability.

3.6 Explainability and Decision-Support Systems

Explainable AI techniques such as Grad-CAM have been applied to verify deep learning predictions in safety-critical applications [18]. However, most pothole detection systems still operate as black-box models and lack dashboard-based decision-support mechanisms required by road authorities [19].

3.7 Summary of Literature and Motivation

From the literature, it is evident that existing approaches emphasize detection accuracy but lack video-level prioritization, degradation awareness, explainability, and decision-support integration [2], [9], [12]. These limitations motivate the proposed video-based, risk-aware, and sensor-free pothole prioritization framework.

4. COMPARATIVE ANALYSIS OF LITERATURE

Author & Year	Technique	Dataset	Key Results	Limitations
Maeda et al. (2018)	CNN-based road damage detection	RDD2018	Accurate pothole and crack detection	Image-level analysis only; no severity or prioritization
Arya et al. (2020)	YOLO-based object detection	RDD	Real-time pothole detection	No severity estimation or temporal aggregation
Zhang et al. (2021)	YOLOv4	Custom road images	High detection mAP	Performance sensitive to lighting and shadows
Polyantseva et al. (2025)	Faster R-CNN + Swin Transformer + PointNet++	Multimodal road data	Improved defect classification using multimodal fusion	High computational complexity; hardware dependency
Qaser et al. (2025)	YOLOv11	Custom dataset (3000+ images)	High accuracy (92.3%) for small potholes	No video-level prioritization or maintenance planning
Shadrach et al. (2025)	YOLOv8 + ResNet	Road images	Improved real-time detection accuracy	Requires edge hardware for deployment
Adebiyi et al. (2025)	Diffusion-based data augmentation (mDDPM)	RDD2022	Significant improvement in training data diversity	Focused only on augmentation, not deployment

Sandhya Devi et al. (2025)	Edge-enhanced YOLOv8	Road images + ESP32-CAM	Robust detection under varying conditions	Hardware-based approach; limited scalability
Pawar et al. (2025)	CNN-based detection + reporting system	Mobile-acquired images	Reliable pothole detection with reporting	No severity grading or risk prioritization
Proposed Work	YOLOv8-Nano + Video-Level Temporal Aggregation	RDD2022 (video frames)	Severity-aware, degradation-aware video-level prioritization	Limited to vision-based analysis (no physical depth sensing)

4. RESEARCH GAPS

Despite significant progress in deep learning–based pothole detection, several critical research gaps remain. Most existing studies focus on frame-level or image-level detection and do not extend analysis to video-level aggregation, which is essential for realistic road inspection workflows . Additionally, current approaches emphasize detection accuracy while neglecting degradation-aware severity modelling, which is necessary for predicting pothole progression and maintenance urgency .

Another major limitation is the reliance on sensor-based or hardware-dependent systems, such as accelerometers, LiDAR, or embedded platforms, which increases deployment cost and restricts scalability across large road networks . Furthermore, many existing systems operate as black-box models with limited explainability and lack integrated decision-support mechanisms required by road authorities .

5. OBJECTIVES OF THE STUDY

The primary objectives of this study are:

- To design and develop a video-based pothole detection and prioritization framework using a lightweight YOLOv8-Nano model .

- To estimate pothole severity and future degradation using visual cues extracted from detected regions of interest .
- To aggregate frame-level pothole information into a video-level road priority score using a risk-aware temporal aggregation strategy .
- To eliminate dependency on physical sensors or IoT hardware by adopting a purely vision-based approach .
- To provide an explainable, dashboard-driven decision-support system for intelligent road maintenance planning .

6. SCOPE OF THE WORK

This work focuses on software-based road condition assessment using road inspection videos and publicly available datasets, particularly RDD2022 . The proposed system operates entirely on visual data and does not incorporate physical sensors, IoT devices, or embedded hardware platforms.

The scope includes pothole detection, severity estimation, degradation prediction, and video-level prioritization within a controlled experimental environment. Real-world field deployment, integration with government maintenance databases, and physical repair execution are beyond the scope of this study but are identified as potential future extensions .

7. PROPOSED METHODOLOGY

The proposed methodology follows a modular end-to-end pipeline aligned with real-world road inspection workflows. Input road videos undergo preprocessing and adaptive frame extraction to ensure computational efficiency . Each extracted frame is processed using YOLOv8-Nano to detect potholes and generate bounding boxes .

Detected regions are analyzed using ROI-based severity estimation based on visual features such as area coverage, edge density, and surface texture roughness . A morphological degradation prediction model simulates future pothole expansion using spatial characteristics . These parameters are combined into a risk score reflecting both current severity and anticipated deterioration

A temporal aggregation engine consolidates frame-level pothole statistics across the video to compute a video-level priority score, enabling actionable maintenance decisions . The results are presented through an interactive dashboard for intuitive visualization and decision support .

8. EXPECTED CONTRIBUTIONS

This work makes the following key contributions:

- A **novel video-based pothole prioritization framework** that extends beyond traditional image-level detection.
- A **risk-aware temporal aggregation model** that converts frame-level detections into road-level maintenance priorities.
- A **sensor-free, vision-only deployment strategy**, improving scalability and cost-effectiveness.
- An **explainable, dashboard-driven decision-support system** suitable for real-world road inspection workflows.
- A comprehensive integration of detection, severity estimation, degradation prediction, and prioritization within a single unified system.

9. CONCLUSION

This paper presented a comprehensive video-based intelligent road condition assessment framework using YOLOv8-Nano. By integrating pothole detection, severity estimation, degradation prediction, and temporal aggregation, the proposed system advances beyond conventional detection-centric approaches .

The framework demonstrates that reliable and scalable road maintenance prioritization can be achieved using only visual data, without reliance on physical sensors or specialized hardware . The experimental results validate the feasibility of deploying a lightweight, explainable, and decision-oriented solution for real-world road infrastructure management. The proposed approach provides a strong foundation for future extensions involving large-scale deployment, multi-city generalization, and integration with automated maintenance scheduling systems .

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